

# Nanoscale Device Engineering (EC 502)



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# Syllabus

## Unit – I

L = 3, T = 0, P = 0, C = 3

- Introduction to Nanoscience and Nanotechnologies

## Unit – 2

- Fabrication Technology & SPICE Device Models

(VLSI Processes: Introduction, Twin-Well CMOS Process, Integrated Devices, MOSFETs, Resistors, Capacitors, pn Junction Diodes, BiCMOS Process, Lateral pnp Transistor, p-Base and Pinched-Base, Resistors, SiGe BiCMOS Process, VLSI Layout, Beyond 20nm Technology Implications of Technology Scaling: Issues in Deep-Submicron Design: Silicon Area, Scaling Implications, Velocity Saturation, Subthreshold Conduction, PVT Variations, Wiring: The Interconnect, SPICE Device Models: The Diode Model, The Zener Diode Model, MOSFET Models, The BJT Model)

## Unit – 3

- Applications of Nanoscience and Nanotechnologies

(Information and Communication Technologies: Integrated Circuits, Data Storage, Photonics, Displays, Information storage devices, Wireless sensing and communication)

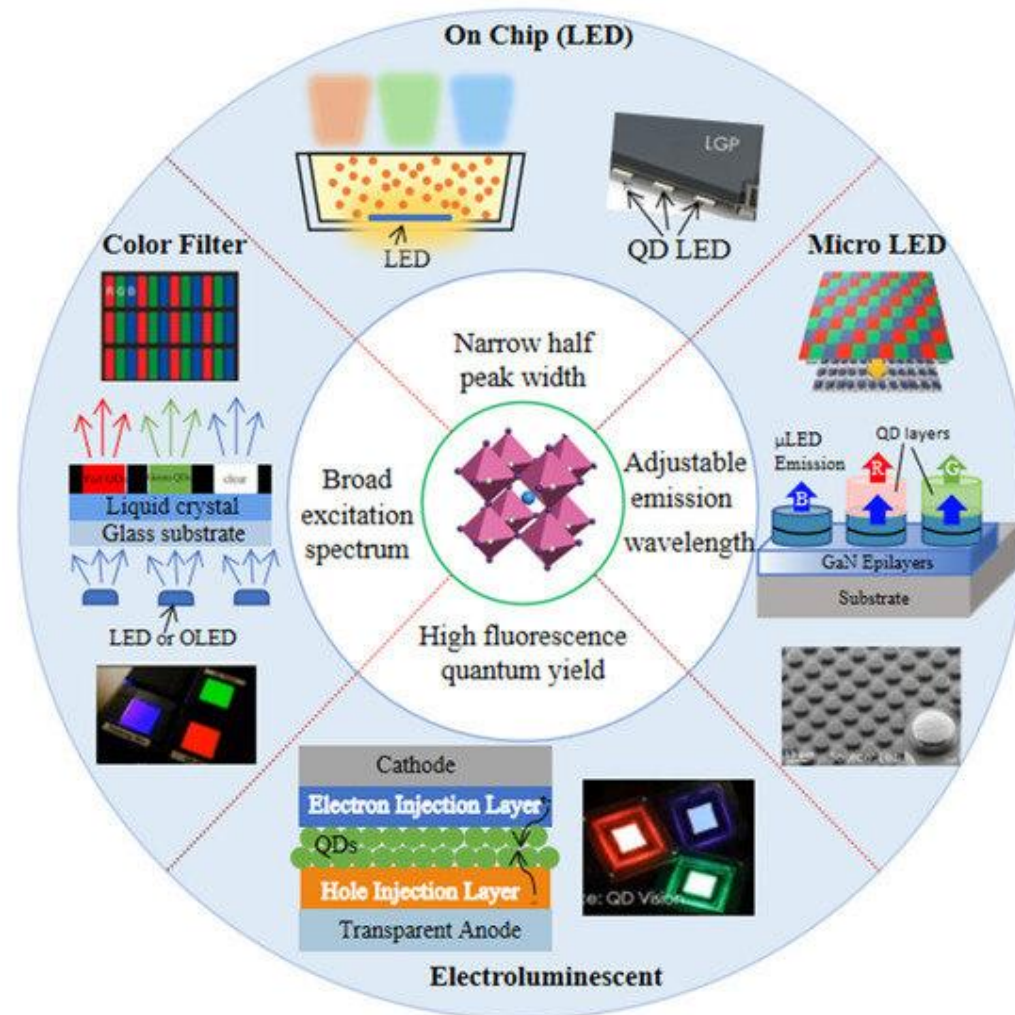
# Displays

Nanoscience enhances display devices through materials like

- **Quantum Dots (QDs)** for brighter and more color-accurate screens,
  - **Carbon NanoTubes (CNTs)** and **Graphene** for flexible, transparent, and more efficient OLEDs and transistors,
  - **Nanoparticles** for advanced touch screens and durable surfaces.
- These advancements allow for features like foldable screens, lower power consumption, and improved durability, impacting consumer electronics like smart phones, TVs, and e-readers.

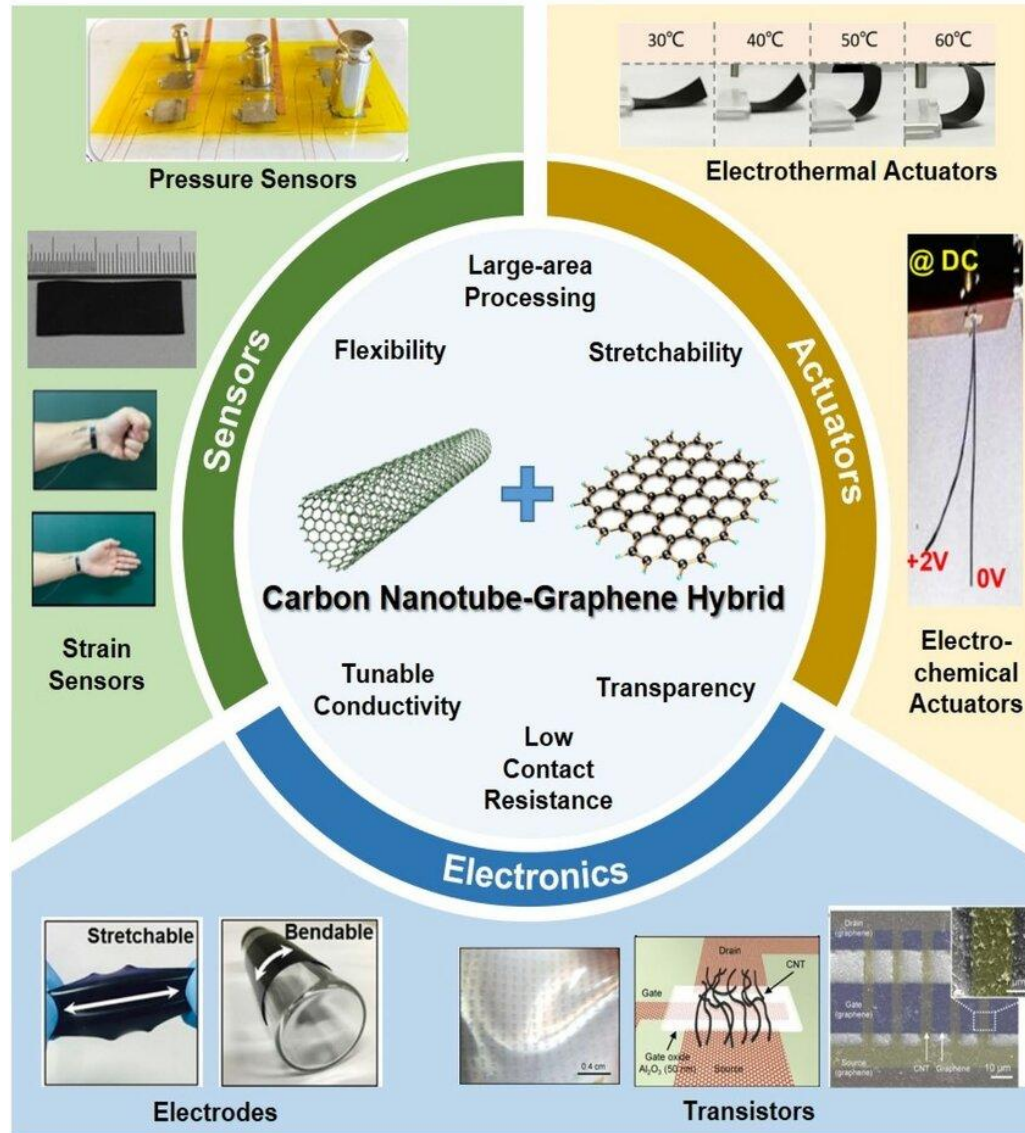
# Displays

## Quantum Dots for Emerging Displays



# Displays

## Carbon Nanotube-Graphene Hybrids for Soft Devices



# Displays

## Key applications

- a. Quantum Dots:** Quantum dots are semiconductor nanocrystals with unique optical properties, such as the ability to emit specific wavelengths of light when excited by an external energy source.
- Quantum dots are increasingly being used in **LED** and **LCD displays** to improve color accuracy, brightness, and efficiency.
  - **Improved Color Performance:** QDs can produce more vibrant colors and have better color stability compared to conventional phosphors in LCDs.
  - **Energy Efficiency:** QDs reduce power consumption by emitting pure light at specific wavelengths, reducing the need for additional filtering layers.
  - **Enhanced Brightness:** Quantum dots improve brightness without sacrificing energy efficiency, making them ideal for high-performance screens such as TVs, monitors, and smartphones.

# Displays

## b. OLEDs and Flexible Displays

- Organic light-emitting diodes (OLEDs) have become a key player in display technology due to their ability to offer vibrant colors, high contrast ratios, and low power consumption. The integration of nanotechnology in OLEDs is improving their performance and enabling the development of **flexible and stretchable displays**.
- **Nanostructured OLEDs:** Nanomaterials such as carbon nanotubes (CNTs) and graphene are being used to enhance the conductivity and efficiency of OLEDs. These materials help create more efficient light-emitting layers and improve the longevity of OLED displays.
- **Flexible and Transparent OLEDs:** By utilizing nanomaterials, OLED displays can be made more flexible and transparent. This technology opens the door to foldable smart phones, rollable televisions, and transparent displays for various applications, such as augmented reality (AR).

# Displays

## c. Nano-Enhanced Touch Screens

- Nanotechnology is also making touch screens more responsive and durable. Nanomaterials like silver nanowires and carbon nanotubes are being incorporated into touch screen technology to improve the conductivity and mechanical properties of the screen.
- **Improved Conductivity:** Silver nanowires are being used as transparent conductive materials, which allows for more efficient touch detection and reduced screen size.
- **Durability:** The use of nanomaterials in touch screens enhances their scratch resistance and overall durability, providing a longer-lasting user experience.

# Displays

## d. Nanotechnology in Display Lighting

- Nanotechnology is being applied to improve the backlighting of displays. For instance, **nano phosphors** are used to enhance the brightness and color rendering of LED backlighting systems.
- **Energy-efficient LED Lighting:** Nano phosphors provide brighter lighting with less energy consumption, contributing to the development of energy-efficient display backlighting systems.
- **Customizable Light Emission:** The tunable properties of nano materials allow for precise control over light emission, making displays more adaptive to different lighting conditions.

# Wireless sensing and communication

Nanotechnology enhances wireless sensing and communication through applications like

- Developing smaller, more sensitive nanosensors for health monitoring and environmental sensing,
- Improving antenna performance with nanomaterials
- Enabling smaller, more efficient wireless components for devices like IoT and wearables.

These advancements lead to faster, more reliable, and energy-efficient wireless networks with applications in medicine, agriculture, and disaster relief.

# Wireless sensing and communication

## Applications in wireless sensing

- **Highly sensitive sensors:** Nanosensors can be integrated with wireless communication to create devices for monitoring air or water quality, soil conditions, and health metrics, say Array Publishers.
- **Real-time health monitoring:** Nanoscale antennas in biosensors can wirelessly transmit real-time patient data, which is critical for monitoring vital signs or disease progression.
- **Precision agriculture:** Wireless nanosensor networks can remotely monitor soil fertility, water levels, and crop health, allowing for more efficient irrigation and fertilizer use.
- **Environmental monitoring:** Nanosensors can be deployed in large networks to monitor for contaminants or track animal movements, and can be used in disaster relief efforts like fire monitoring.

# Wireless sensing and communication

## Applications in wireless communication

- **Improved antennas:** Nanomaterials can be used to design smaller, more efficient antennas with better performance, precision, and power efficiency for a wide range of devices.
- **Miniaturized components:** Nanotechnology enables the creation of smaller, lower-power integrated circuits, enabling more compact wireless devices.
- **Faster and more efficient networks:** Nanosystems are crucial for future wireless technologies like 5G and beyond, promising higher speeds and lower interference.
- **Advanced data storage:** Nanotechnology introduces new methods for data storage within communication systems, such as using nanocrystal storage medias.

# Wireless sensing and communication

## Future potential

- **Increased efficiency and performance:** Nanomaterials allow for precise control of antenna properties, leading to greater efficiency and tailored performance for specific applications.
- **Energy efficiency:** The development of tiny, energy-harvesting devices powered by sources like light or heat, combined with low-power sensors, is a major benefit of nanotechnology.
- **Cost reduction:** While initial research and development costs are high, manufacturing costs can decrease with mass production of components like printed nanoparticle inks for sensors.



*Thank You*